Fixed occupations and sharp collapse thresholds

An exact one-dimensional result and a two-dimensional Lieb–Thirring problem

Shuxing Yang

9 October 2026

Abstract

For common-density nonlinear Schrödinger systems, the conserved Gram spectrum determines an occupation-dependent universal collapse threshold. In two dimensions we give the exact weighted spectral reformulation, prove monotonicity under majorization, and record explicit bounds. In the one-dimensional quintic system the sharp inequality of Read–Schulz yields the exact strict-side threshold for every finite occupation spectrum. The two-dimensional two-state sharp constant remains undetermined. These statements concern uniform protection below threshold and existence of blowup above it; they do not classify every solution at equality.

1 The two-dimensional threshold

Fix a finite probability vector $p_1 \geq \cdots \geq p_r > 0$, $\sum_j p_j = 1$, and consider

$$i\partial_t u_j = -\Delta u_j - \rho u_j, \qquad \rho = \sum_{j=1}^{r} |u_j|^2, \qquad x \in \mathbb{R}^2. \tag{1}$$

For $U = (u_1, \ldots, u_r)^T$, write

$$\Gamma(U) = \int UU^*, \quad M = \operatorname{Tr}\Gamma, \quad A(U) = \sum_j \|\nabla u_j\|_2^2, \quad F(U) = \int \rho^2, \quad E = \tfrac{1}{2}A - \tfrac{1}{4}F.$$

The entire matrix Γ is conserved. Indeed the common self-adjoint operator $H_U = -\Delta - \rho$ gives

$$\frac{d}{dt}\int u_j\overline{u_k} = -i\int (H_U u_j)\overline{u_k} + i\int u_j\overline{H_U u_k} = 0.$$

Approximation extends this identity to the finite-component H^1 flow. The usual local theory and blowup alternative apply; the cubic nonlinearity has derivative bounded by $3|U|^2|V|$.

For an L^2-orthonormal tuple $\Phi = (\phi_1, \ldots, \phi_r)$, define

$$\rho_p = \sum_j p_j|\phi_j|^2, \quad A_p = \sum_j p_j\|\nabla\phi_j\|_2^2, \quad C(p) = \sup_{\Phi} \frac{\int \rho_p^2}{A_p}, \qquad M_*(p) = \frac{2}{C(p)}.$$

The constant is positive and finite: $\|\sqrt{\rho_p}\|_2 = 1$ and $\|\nabla\sqrt{\rho_p}\|_2^2 \leq A_p$ reduce finiteness to scalar interpolation. With $u_j = \sqrt{Mp_j}\phi_j$,

$$F(U) \leq MC(p)A(U) \quad \text{when } \Gamma(U) = M\,\mathrm{diag}(p). \tag{2}$$

A constant internal unitary transformation covers any Gram matrix with the same spectrum.

Theorem 1 (Strict-side dynamical meaning)**.** *If $M < M_*(p)$, all H^1 data with Gram matrix $M\,\mathrm{diag}(p)$ generate global solutions uniformly bounded in H^1 in both time directions. If $M > M_*(p)$, there exist smooth compactly supported data with exactly this Gram matrix whose solutions blow up in finite time in both directions.*

Proof. Conservation and (2) imply

$$E(U_0) \geq \tfrac{1}{2}(1 - M/M_*)A(U(t)), \qquad A(U(t)) \leq \frac{2E(U_0)}{1 - M/M_*}.$$

Mass conservation and continuation give the subthreshold conclusion. Above threshold choose an orthonormal tuple with $\int \rho_p^2/A_p > 2/M$. Approximate it in H^1 by a smooth compactly supported vector F_n. Its Gram matrix $G_n = \int F_n F_n^*$ tends to I, and $G_n^{-1/2}F_n$ is exactly orthonormal and converges in H^1. The L^4 embedding preserves the strict quotient inequality. Multiplying the components by $\sqrt{Mp_j}$ gives the prescribed Gram matrix and $E < 0$. The variance satisfies

$$I(t) = \int |x|^2 \rho(t,x), \quad I'(t) = 4\,\mathrm{Im} \sum_j \int x \cdot \nabla u_j \overline{u_j}, \quad I'' = 8A - 4F = 16E.$$

Thus $I(t) = I(0) + tI'(0) + 8t^2E$ cannot remain nonnegative for all positive or all negative times. □

No maximizer is needed. The superthreshold assertion is existential.

2 The weighted spectral problem

For $0 \neq V \geq 0$ in $L^2(\mathbb{R}^2)$, let $e_1(V) \geq e_2(V) \geq \cdots \geq 0$ be the magnitudes of the negative eigenvalues of $-\Delta - V$, padded by zeros. Put

$$K_p(V) = \sum_j p_j e_j(V), \qquad L(p) = \sup_V \frac{K_p(V)}{\int V^2}.$$

Weighted min–max gives $K_p(V) = \sup_\Phi (\int V\rho_p - A_p)$. Diffuse states at infinity supply the zero-filled entries. This is the weighted form of the established kinetic/spectral duality [2, 3].

Proposition 2 (Normalization)**.**

$$C(p) = 4L(p), \qquad M_*(p) = \frac{1}{2L(p)}.$$

Proof. For $X = \int \rho_p^2$ and $A = A_p$, take $V = b\rho_p$ with $b = 2A/X$. Then $L(p) \geq (bX - A)/(b^2X) = X/(4A)$. Conversely,

$$\int V\rho_p - A \leq \|V\|_2 \sqrt{C(p)A} - A \leq \tfrac{1}{4}C(p)\|V\|_2^2.$$

Take the relevant suprema. □

3 Redistribution of occupations

If $p \succ q$ in majorization order, Abel summation against the decreasing sequence $e_j(V)$ gives $\sum_j p_j e_j(V) \geq \sum_j q_j e_j(V)$. Hence

$$p \succ q \quad \Longrightarrow \quad M_*(p) \leq M_*(q). \tag{3}$$

More evenly distributed occupations never lower the universal non-collapse threshold. Extended by permutation symmetry, $L(p)$ is convex, being a supremum of linear functions of p. Thus $1/M_*(p)$ is convex.

Let $M_{\mathrm{c}} = \|Q\|_2^2$ for $-\Delta Q + Q - Q^3 = 0$ in $\mathbb{R}^2$, so $\int |f|^4 \le (2/M_{\mathrm{c}})\|f\|_2^2\|\nabla f\|_2^2$. Define

$$L_{1,2}^{(k)} = \sup_{V\ge 0, V\neq 0} \frac{\sum_{j=1}^k e_j(V)}{\int V^2}.$$

Then $L_{1,2}^{(1)} = 1/(2M_{\mathrm{c}})$ and $M_*(1/k,\dots,1/k) = k/(2L_{1,2}^{(k)})$. For two occupations,

$$p_1 L_{1,2}^{(1)} \le L(p) \le (p_1 - p_2)L_{1,2}^{(1)} + p_2 L_{1,2}^{(2)}.$$

Consequently,

$$L_{1,2}^{(2)} = L_{1,2}^{(1)} \quad\Longleftrightarrow\quad M_*(p_1,p_2) = \frac{M_{\mathrm{c}}}{\max(p_1,p_2)} \quad \text{for all } p.$$

The reverse implication already follows from $p_1 = p_2 = 1/2$.

Read–Schulz [1, Theorem 2] give $L_{1,2} \le 1/(4\sqrt{3}\pi)$. Since $K_p \le p_1 \sum_j e_j$, this yields $M_*(p) \ge 2\sqrt{3}\pi/p_1$. The scalar-projection estimate from the preceding occupation derivation gives $C(p) \le (1+p_1)/M_{\mathrm{c}}$. Together,

$$\max\left\{\frac{2M_{\mathrm{c}}}{1+p_1}, \frac{2\sqrt{3}\pi}{p_1}\right\} \le M_*(p) \le \frac{M_{\mathrm{c}}}{p_1}. \tag{4}$$

In particular $4\sqrt{3}\pi \le M_*(1/2,1/2) \le 2M_{\mathrm{c}}$. The stronger spectral lower bound uses the external 2026 theorem.

4 Exact one-dimensional thresholds

Consider instead the one-dimensional quintic system

$$i\partial_t u_j = -\partial_x^2 u_j - \rho^2 u_j, \qquad \rho = \sum_j |u_j|^2, \qquad E = \tfrac{1}{2}A - \tfrac{1}{6}F, \quad A = \sum_j \|u_j'\|_2^2, \quad F = \int \rho^3.$$

Its Gram matrix is conserved by the same argument. Here

$$Q(x) = 3^{1/4}\operatorname{sech}^{1/2}(2x), \qquad -Q'' + Q - Q^5 = 0, \qquad M_{\mathrm{c}}^{(1)} = \|Q\|_2^2 = \frac{\sqrt{3}\pi}{2}.$$

The external orthonormal inequality of Read–Schulz, including its complex formulation, is

$$\int_{\mathbb{R}} \left(\sum_{j=1}^k |\phi_j|^2\right)^3 \le \frac{4}{\pi^2}\sum_{j=1}^k \|\phi_j'\|_2^2. \tag{5}$$

Theorem 3. *For every finite occupation spectrum, the strict-side universal threshold is*

$$M_*^{(1)}(p) = \frac{\sqrt{3}\pi}{2p_1} = \frac{M_{\mathrm{c}}^{(1)}}{p_1}.$$

Below it every prescribed-Gram H^1 solution is global and uniformly bounded. Above it smooth compactly supported prescribed-Gram data exist with finite-time blowup in both directions.

Proof. Diagonalize Γ and write $u_j = \sqrt{\lambda_j}\phi_j$, with $\mu = \lambda_1 = Mp_1$. For $0 < t < \mu$, set $\rho_t = \sum_{\lambda_j > t} |\phi_j|^2$ and $A_t = \sum_{\lambda_j > t} \|\phi_j'\|_2^2$. Layer integration gives $\rho = \int_0^\mu \rho_t\, dt$ and $A = \int_0^\mu A_t\, dt$. Jensen and (5) imply

$$F \le \mu^2 \int_0^\mu \int \rho_t^3\, dx\, dt \le \frac{4\mu^2}{\pi^2} A = 3\left(\frac{\mu}{M_{\mathrm{c}}^{(1)}}\right)^2 A.$$

Thus $E \ge \frac{1}{2}[1 - (\mu/M_{\mathrm{c}}^{(1)})^2]A$ proves the subthreshold claim. For sharpness, approximate $Q/\sqrt{M_{\mathrm{c}}^{(1)}}$ by a normalized smooth compactly supported function. Choose the other normalized smooth modes on disjoint distant intervals and dilate them so that their kinetic energies tend to zero. Orthogonality is exact and

$$\frac{\int (\sum_j p_j |\phi_j|^2)^3}{\sum_j p_j \|\phi_j'\|_2^2} \longrightarrow \frac{4p_1^2}{\pi^2}.$$

If $Mp_1 > M_{\mathrm{c}}^{(1)}$, a finite construction has $E < 0$ and finite variance. The critical identity $I'' = 8A - \frac{8}{3}F = 16E$ gives blowup as in Theorem 1. □

5 The planar endpoint and optical normalization

The remaining two-state target is

$$e_1(V) + e_2(V) \overset{?}{\le} \frac{1}{2M_{\mathrm{c}}} \int_{\mathbb{R}^2} V^2.$$

Frank–Gontier–Lewin's strict binding range is $\gamma > \max\{0, 2 - d/2\}$, which at $d = 2$ does not include $\gamma = 1$. Their finite-rank optimizer theory does not itself determine this endpoint [3, 4]. If the target holds, every two-mode threshold is M_{c}/p_1; if it fails, the equal-mode threshold is strictly below $2M_{\mathrm{c}}$.

For the common-density optical equation

$$i\partial_z a_j = -D\Delta_x a_j - g\left(\sum_k |a_k|^2\right) a_j, \qquad D, g > 0, \quad x \in \mathbb{R}^2,$$

put $x = \ell y$, $z = \ell^2 t/D$ and $a_j = \sqrt{D/(g\ell^2)}\, u_j$. Physical total power satisfies

$$P = \int \sum_j |a_j|^2 = \frac{D}{g} M, \qquad P_*(p) = \frac{D}{g} M_*(p).$$

The two-state target therefore asks whether two equally occupied orthogonal modes double the infimum of powers admitting collapse. The model requires common scalar dispersion and common density coupling. Coherence-dependent collapse has established optical antecedents [7]. In the one-dimensional quintic model the corresponding normalization is $P = \sqrt{D/g}\, M$, giving $P_* = (M_{\mathrm{c}}^{(1)}/p_1)\sqrt{D/g}$.

References

[1] L. Read and M. R. Schulz, *The sharp one-dimensional Lieb–Thirring inequality for the sum of eigenvalues*, arXiv:2609.10478v2 (2026). https://arxiv.org/abs/2609.10478v2.

[2] J. Dolbeault, P. Felmer, M. Loss and E. Paturel, *Lieb–Thirring type inequalities and Gagliardo–Nirenberg inequalities for systems*, J. Funct. Anal. 238 (2006), 193–220. https://arxiv.org/abs/math-ph/0506052.

[3] R. L. Frank, D. Gontier and M. Lewin, *The nonlinear Schrödinger equation for orthonormal functions II: Application to Lieb–Thirring inequalities*, Comm. Math. Phys. 384 (2021), 1783–1828. https://arxiv.org/abs/2002.04964.

[4] R. L. Frank, D. Gontier and M. Lewin, *Optimizers for the finite-rank Lieb–Thirring inequality*. https://arxiv.org/abs/2109.05984.

[5] G. K. Duong, T. M. T. Le, P. T. Nam and P.-T. Nguyen, *Finite-rank optimizers for the mass-supercritical Lieb–Thirring and Hardy–Lieb–Thirring inequalities*. https://arxiv.org/abs/2510.24148.

[6] R. Killip and M. Visan, *Nonlinear Schrödinger equations at critical regularity*, Clay Mathematics Proceedings 17 (2013). https://www.math.ucla.edu/~visan/ClayLectureNotes.pdf.

[7] O. Bang, D. Edmundson and W. Królikowski, *Collapse of incoherent light beams in inertial bulk Kerr media*, Phys. Rev. Lett. 83 (1999), 5479–5482. https://doi.org/10.1103/PhysRevLett.83.5479.